Reference Selection Biases in Retrieval Augmented Generation for Scientific Texts

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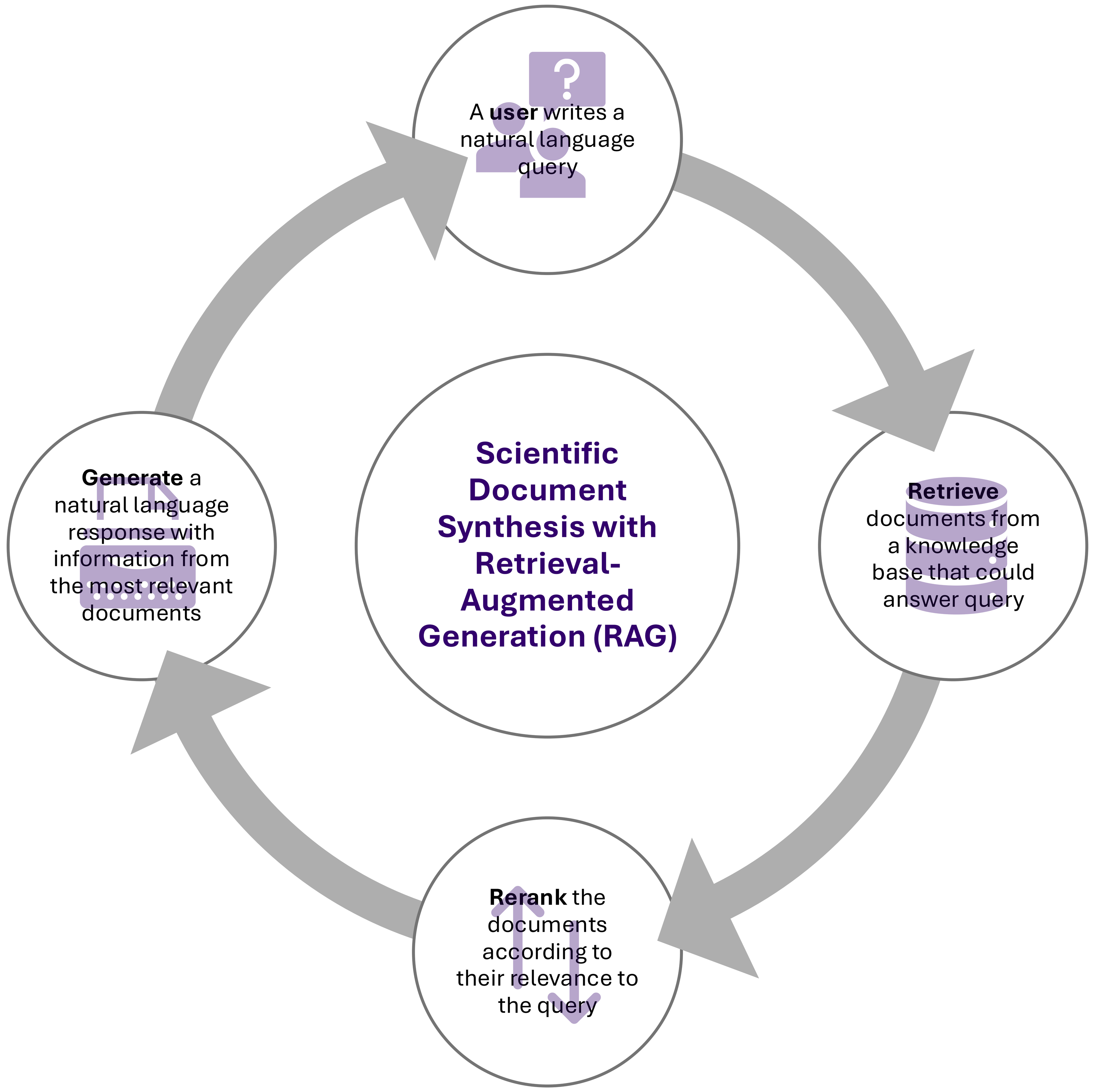
CRA-WP Computing Research Association Widening Participation

Background and Research Question

- When choosing which works to cite, people have biases like preferring newer works [1], works with more citations [2], or works that are open access [3].
- How does using retrieval augmented generation (RAG) change citation biases when writing Related Works sections?



Data and Approach

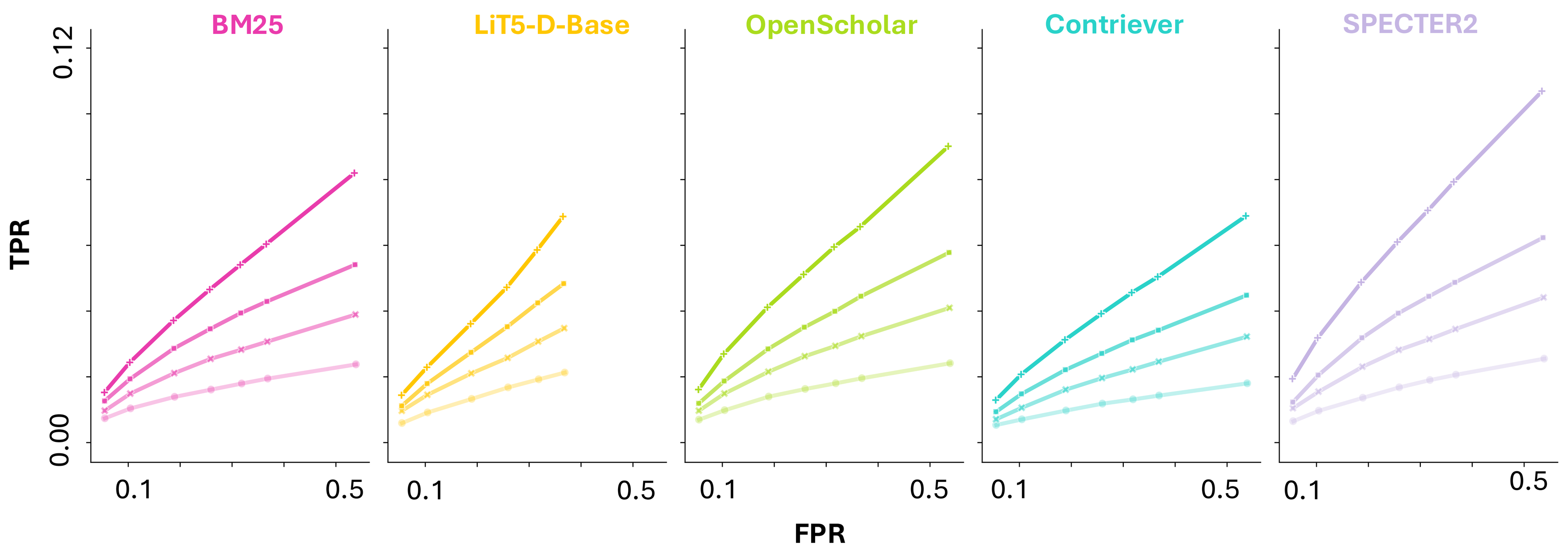
- **Queries:** ~9,000 titles and abstracts from CORWA [4], a dataset of published NLP papers with full-text Related Works sections and annotated inline citations
- **Index**: ~27,000 NLP publications venues which have titles and abstracts in Semantic Scholar [5]
- **Procedure:** Compare the metadata features of publications selected using RAG to those in the ground-truth Related Works sections of the query papers

Which biases do retrievers and rerankers exhibit?

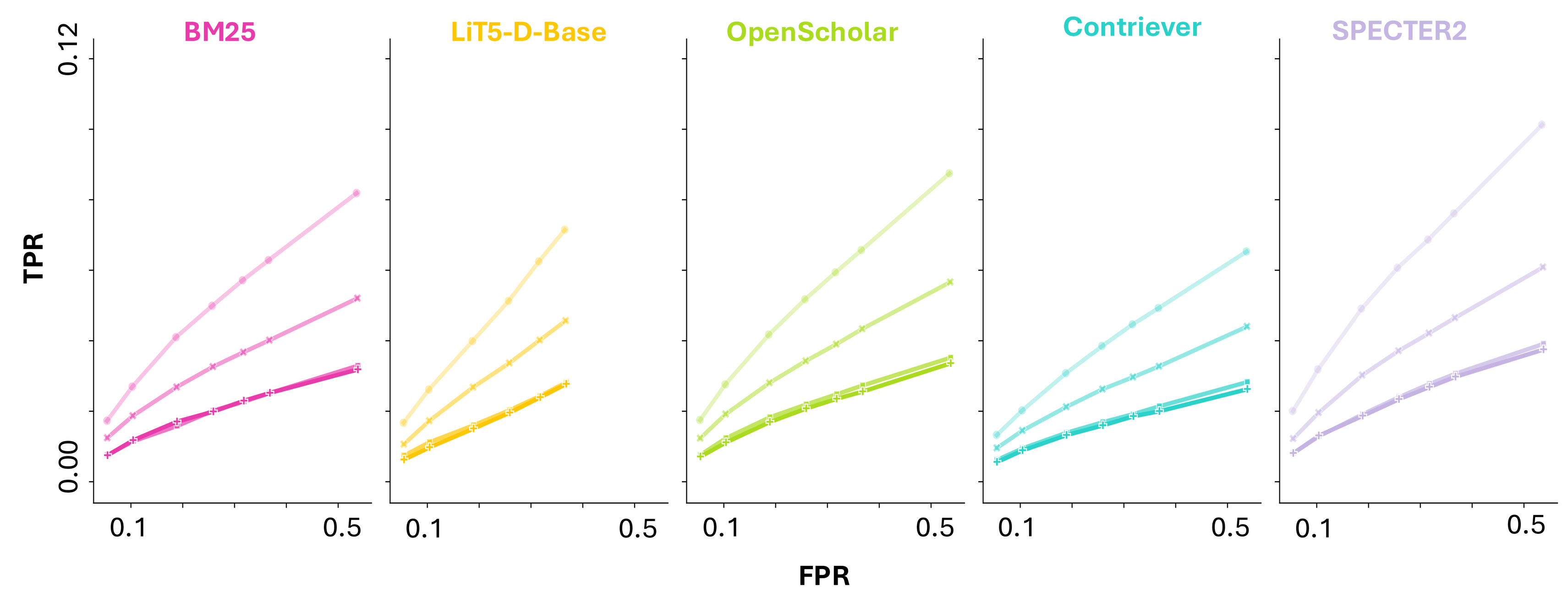
Correct papers are more likely to be selected if they have...

1. Higher numbers of citations
2. Fewer authors
3. Been published for a while

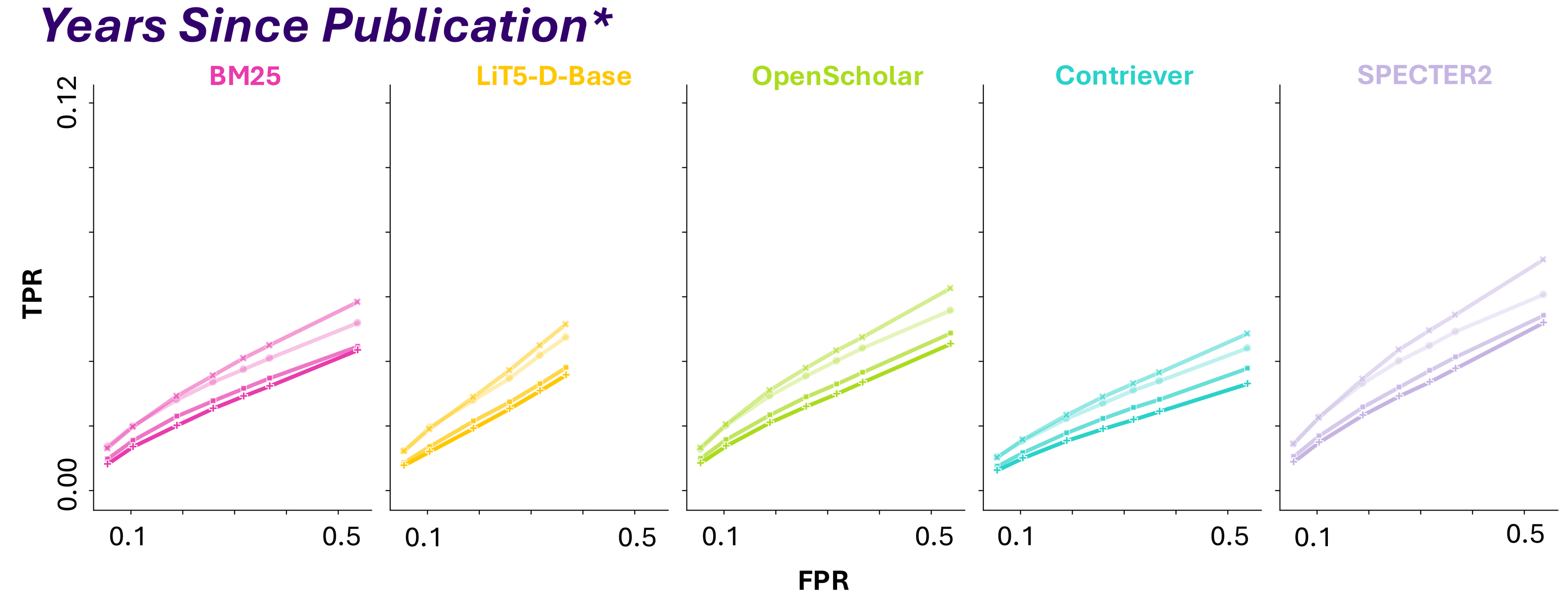
*Number of Citations**



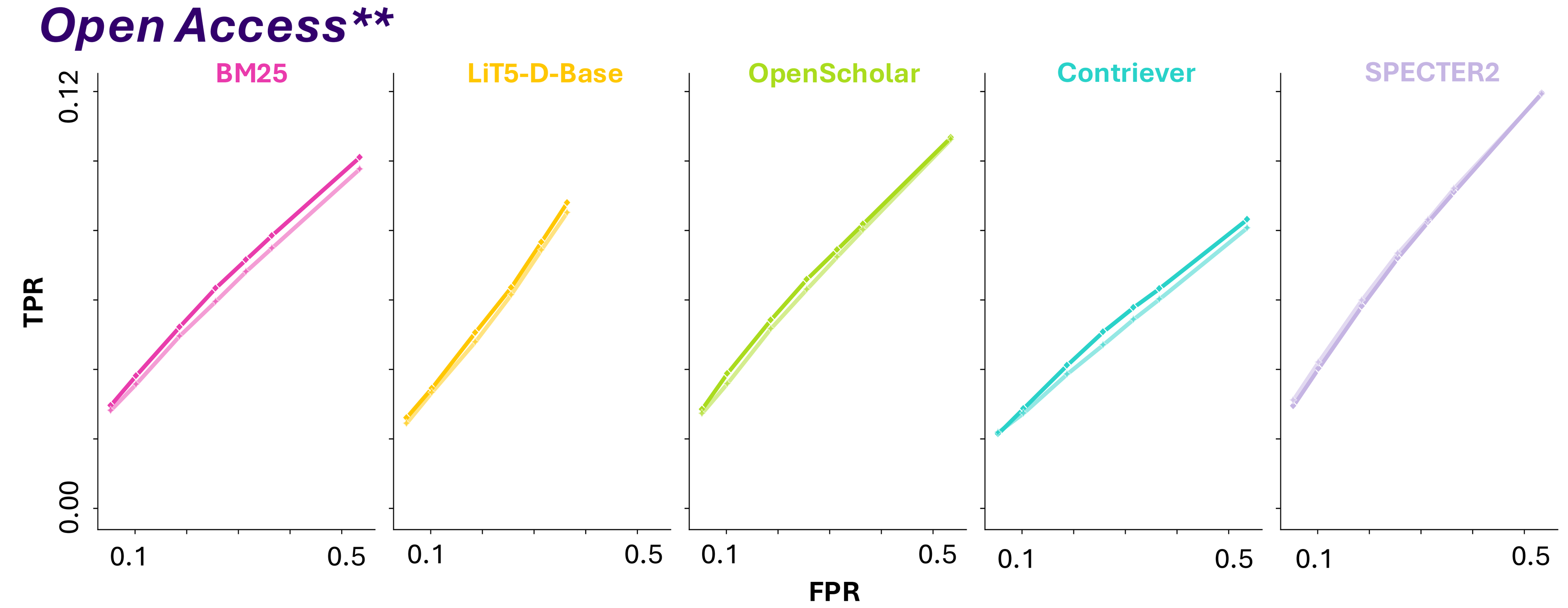
*Number of Authors**



*Years Since Publication**



*Open Access***



* Darker colors represent quartiles of increasing values for continuous variables

** Darker colors represent papers that are open access; lighter values represent those that are not

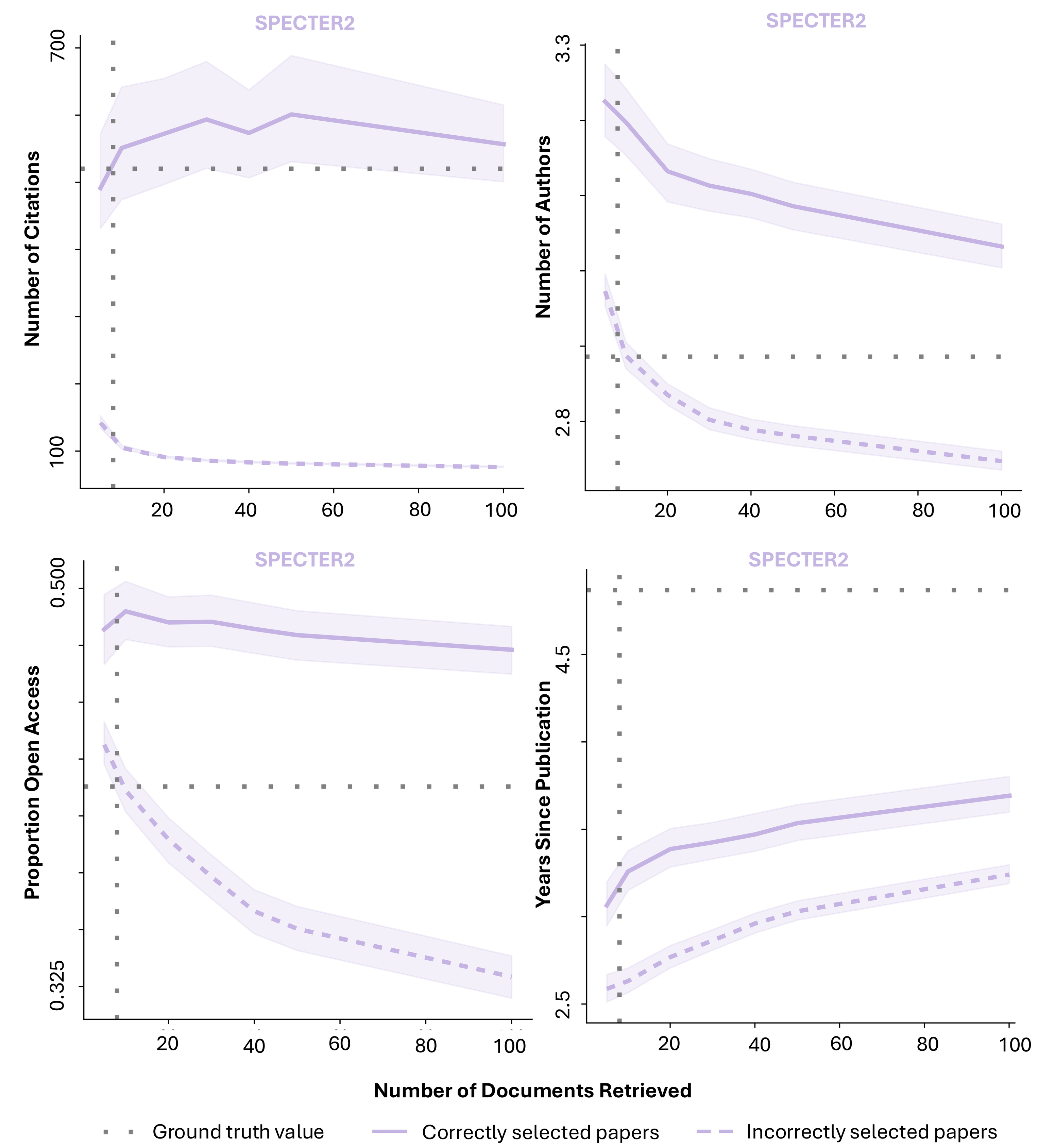
How do references selected in RAG compare to the references selected by authors?

Correct papers have...

1. Similar # of citations
2. More authors
3. More open access
4. Been published more recently

Incorrect papers have...

1. Lower # of citations
2. Fewer authors
3. Less open access
4. Been published more recently



Future Work

1. Include more state-of-the-art rerankers in analyses
2. Include other features which are known to affect citation rates, including prestige of publication venues and author affiliations
3. Study usage of selected references in generated text

References

[1] Wahle, J. P., Ruas, T., Abdalla, M., Gipp, B., & Mohammad, S. M. (2024). Citation Amnesia: On The Recency Bias of NLP and Other Academic Fields. *arXiv preprint arXiv:2402.12046*. **[2]** Algaba, A., Mazijn, C., Holst, V., Tori, F., Wenmackers, S., & Ginis, V. (2024). Large language models reflect human citation patterns with a heightened citation bias. *arXiv preprint arXiv:2405.15739*. **[3]** Eysenbach, G. (2006). Citation advantage of open access articles. *PLoS biology*, *4*(5), e157. **[4]** Li, X., Mandal, B., & Ouyang, J. (2022, July). CORWA: A Citation-Oriented Related Work Annotation Dataset. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies* (pp. 5426-5440). **[5]** Kinney, R., Anastasiades, C., Authur, R., Beltagy, I., Bragg, J., Buraczynski, A., ... & Weld, D. S. (2023). The semantic scholar open data platform. *arXiv preprint arXiv:2301.10140*.